

Fig. 2: Actual-size PCB layout for the high-impedance audio buffer with JFET

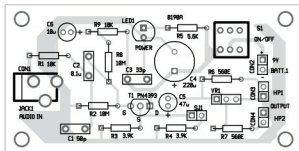


Fig. 3: Component layout of the PCB

power consumption. Bias depends on the input signal, parameters of the JFET and power supply.

JFETs have wide tolerances, which is the main issue during their application. Fortunately, this is not a problem here. If a single circuit is required, choose an appropriate value for R3 depending on the JFET. In that case, SJ1 and R4 can be omitted. If some other JFET devices are required, connect R4 to source S of the JFET through SJ1 without changing the value of R3 to adjust the circuit to the parameters of the JFET.

This buffer has two outputs available at connectors CON3 and CON4. The outputs can drive loads of 10-kilo-ohm or more, simultaneously. Higher impedance loads are preferred. The loads can be reduced to 2-kilo-ohm without overloading the circuit, but amplitude of the signal will drop. Usually, that is not a problem. The circuit can drive two high-impedance 2-kilo-ohm headphones (HP1 and HP2) connected to CON3 or CON4 (a total load of 4-kilo-ohm).

Signal level on output CON3 is not adjustable but signal level on

CON4 is, using pot-meter VR1.

Choice of power supply depends on the peak-to-peak amplitude from the signal source. One or two 9V batteries of type 6F22 can be used. These provide 9V or 18V of power supply voltage, covering practically all passive pickups for the musical instruments and other high-impedance sensors. Some JFETs allow higher power supply. A well-filtered DC wall adaptor can also be used.

Construction and testing

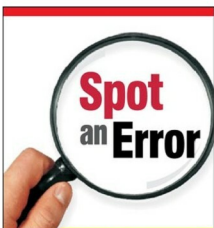
An actual-size, single-side PCB for the high-impedance audio buffer with JFET is shown in Fig. 2 and its component layout in Fig. 3.

This circuit may require appropriate selection of values of resistors R3 and R4 in order to optimise the peak-to-peak amplitude of the output signal. It is appropriate for battery operation, which is important for portable applications.

The circuit can be mounted in a small box near the pickup or attached to the strap of the string instrument. Due to its high input impedance, input cable should be shorter than one metre and shielded or, at least, twisted. The gain is lower than unity but that is not a problem because most pickups provide strong signals. **EFY**



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